

ModuleDict#

<https://pytorch.org/docs/stable/generated/torch.nn.ModuleDict.html>

Description:

Holds submodules in a dictionary. ModuleDict can be indexed like a regular Python dictionary, but modules it contains are properly registered, and will be visible by all Module methods. ModuleDict is an ordered dictionary that respects the order of insertion, and in `update()`, the order of the merged `OrderedDict`, `dict` (started from Python 3.6) or another `ModuleDict` (the argument to `update()`).

Function Signature

```
class torch.nn.ModuleDict(modules=None)[source]#
```

Parameters

```
clear()[source]#
```

```
items()[source]#
```

Return type

Returns:

ValuesView[Module]

Code Examples

```
class MyModule(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.choices = nn.ModuleDict(
            {"conv": nn.Conv2d(10, 10, 3), "pool":
nn.MaxPool2d(3)}
        )
        self.activations = nn.ModuleDict(
            [["lrelu", nn.LeakyReLU()], ["prelu", nn.PReLU()]]
        )

    def forward(self, x, choice, act):
        x = self.choices[choice](x)
        x = self.activations[act](x)
        return x
```

Notes & Warnings

NOTE If modules is an OrderedDict, a ModuleDict, or an iterable of key-value pairs, the order of new elements in it is preserved.

Detailed Documentation

Documentation

Holds submodules in a dictionary.

ModuleDict can be indexed like a regular Python dictionary, but modules it contains are properly registered, and will be visible by all Module methods.

ModuleDict is an ordered dictionary that respects

the order of insertion, and

in `update()`, the order of the merged `OrderedDict`, `dict` (started from Python 3.6) or another `ModuleDict` (the argument to `update()`).

Note that `update()` with other unordered mapping types (e.g., Python's plain `dict` before Python version 3.6) does not preserve the order of the merged mapping.

`modules(iterable, optional)` – a mapping (dictionary) of (string: module) or an iterable of key-value pairs of type (string, module)

Remove all items from the `ModuleDict`.

Return an iterable of the `ModuleDict` key/value pairs.

`ItemsView[str, Module]`

Return an iterable of the `ModuleDict` keys.

Remove key from the `ModuleDict` and return its module.

key (str) – key to pop from the ModuleDict

Update the ModuleDict with key-value pairs from a mapping, overwriting existing keys.

If modules is an OrderedDict, a ModuleDict, or an iterable of key-value pairs, the order of new elements in it is preserved.

modules (iterable) – a mapping (dictionary) from string to Module, or an iterable of key-value pairs of type (string, Module)

Return an iterable of the ModuleDict values.